

BASIC CONSTANTS

Mathematical Constants		
π	3.141592653589793	
e	2.718281828459045	
Physical Constants		
Speed of light (c)	299,792,458	m/s
Constant of gravitation (G)	6.67259×10^{-11}	Nm^2/kg^2
Universal gas constant (R)	8,314.4621	J/kmol-K
Stefan-Boltzmann constant (σ)	5.670373×10^{-8}	$W/m^2\text{-}K^4$
Acceleration of gravity (g)	9.80665	m/s^2
Standard atmosphere, sea level	101,325	Pa
Astronomical Constants		
Astronomical unit (AU)	149,597,870	km
Light year (ly)	9.460530×10^{12}	km
Parsec (pc)	3.261633	ly
Sidereal year	365.256366	days
Mass of Sun	1.9891×10^{30}	kg
Radius of Sun	696,000	km
Mass of Earth	5.9737×10^{24}	kg
Equatorial radius of Earth	6,378.137	km
Earth oblateness	1/298.257	
Obliquity of the ecliptic, epoch 2000	23.4392911	degrees
Mean lunar distance	384,403	km
Radius of Moon	1,738	km
Mass of Moon	7.348×10^{22}	kg
Luminosity of Sun	3.839×10^{26}	W
Solar constant, at 1 AU	1,366	W/m^2
Solar maxima	1990 + 11n	(date)
Spaceflight Constants		
GM (Sun)	$1.32712438 \times 10^{20}$	m^3/s^2
GM (Earth)	3.986005×10^{14}	m^3/s^2

GM (Moon)	4.902794×10^{12}	m^3/s^2
GM (Mars)	4.282831×10^{13}	m^3/s^2
J_2 (Earth)	0.00108263	
J_2 (Moon)	0.0002027	
J_2 (Mars)	0.00196045	

Planet Data

Name	Equatorial Radius (km)	Mass (10^{24} kg)	Surface Gravity (m/s^2)	GM (10^{15} m^3/s^2)	Semimajor Axis (10^6 km)	Eccentricity	Orbit Inclination (degrees)	Sidereal Period (days)	Sidereal Rotation Period (hours)
Mercury	2,439.7	0.3302	3.70	0.02203	57.91	0.2056	7.00	87.960	1,407.6
Venus	6,051.8	4.8685	8.87	0.3249	108.21	0.0067	3.39	224.701	-5,832.5
Earth	6,378.1	5.9742	9.80	0.3986	149.60	0.0167	0.000	365.256	23.9345
Mars	3,397.0	0.64185	3.71	0.04283	227.92	0.0935	1.850	686.980	24.6229
Jupiter	71,492	1,898.6	24.79	126.686	778.57	0.0489	1.304	4,332.59	9.9250
Saturn	60,268	568.46	10.44	37.931	1,433.53	0.0565	2.485	10,759.2	10.656
Uranus	25,559	86.832	8.87	5.794	2,872.46	0.0457	0.772	30,685.4	-17.24
Neptune	24,764	102.43	11.15	6.835	4,495.06	0.0113	1.769	60,189	16.11
Pluto	1,195	0.0125	0.58	0.00083	5,869.66	0.2444	17.16	90,465	-153.29

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